Peak Prediction Using Quantile Regression and Asymmetric Laplace Noise

Abstract

Predicting the location of peaks in time series is important when decisions must be made before the peak is fully observed. We model the observations as a Gaussian trend with additive noise and develop a quantile-based procedure for estimating its amplitude, location, and spread. For fixed location and spread, the amplitude estimator is a weighted τ-quantile of a transformed version of the observations. We introduce a τ-aligned asymmetric Laplace noise model whose negative log-likelihood is proportional to the quantile check loss, providing a probabilistic interpretation of the criterion and allowing the effects of the quantile level τ and the signal-to-noise ratio to be studied in a unified framework. When the estimated peak location does not stabilize early enough to yield an accurate point prediction, we propose a two-stage construction based on the oscillations of the sequential estimates. It produces candidate entry and exit times, and hence candidate peak regions, using only the estimation trajectory and not the true peak location. The observed peak can be used retrospectively to evaluate the candidates. The methodology is illustrated on daily COVID-19 cases in China, France, and Germany.

Keywords: Peak prediction; quantile regression; asymmetric Laplace noise; peak-region entry and exit times; time series; COVID-19.

Mathematics subject classification: 62M10, 62G35, 60G35.
JEL classification: C22, C21, C53, I18.

1 Introduction

Predicting peaks in time series data is important across a wide range of applications involving proactive decision-making, risk management, resource allocation, and disaster preparedness, such as anticipating disease outbreaks in epidemiology or peak demand in energy management. Peak prediction is challenging because observations are affected by noise, and the underlying signal may exhibit nonlinear or otherwise complex behavior. Measurement errors and random fluctuations can make the location and amplitude of a peak difficult to identify from partial observations, and real-world time series may contain multiple peaks, irregular peak shapes, or other temporal structures that are not adequately described by simple forecasting models.

Popular forecasting approaches such as Autoregressive Integrated Moving Average (ARIMA) models and Recurrent Neural Networks (RNNs) can be effective for describing temporal dependence and forecasting general patterns, but they are not specifically designed to estimate the timing and magnitude of a future peak. A model that provides a good overall forecast may not necessarily provide an accurate estimate of the location of a future peak. There has also been comparatively little theoretical work devoted specifically to peak prediction in time series; one contribution in this direction is the thesis of Tucker Tucker (2020), which studies the interplay between trend, seasonality, and noise; see also Pei and Shaman (2017), Yu (2019).

Our approach builds on the framework developed in Dermoune et al. (2025). We consider observations available at an observation horizon T and a prescribed future horizon $h > T$. The observations are modeled, as in Section 2, as a Gaussian trend of amplitude $A > 0$, peak location L, and spread $S > 0$, corrupted by additive noise. The objective is to estimate (A, L, S) from the observations available up to time T and, in particular, to predict the location L of the peak before it is fully observed.

Our first contribution is to replace the least-absolute-deviation criterion used in the symmetric case by a general quantile regression criterion indexed by $\tau \in (0, 1)$. As we show in Section 2, for fixed location and spread, the amplitude minimizing this criterion is a weighted τ-quantile of a transformed version of the observations; the weighted median of the symmetric case is recovered at $\tau = 1/2$.

Our second contribution is a probabilistic interpretation of the criterion. In Section 4 we introduce a τ-aligned asymmetric Laplace noise model whose negative log-likelihood is proportional to the quantile check loss. This allows the effects of the quantile level τ and of the signal-to-noise ratio to be studied within a unified framework, and it recovers the symmetric Laplace model at $\tau = 1/2$. Sections 5 and 6 examine the resulting estimated peak-location trajectories under high and low signal-to-noise ratios.

Our third contribution concerns the situation in which the estimated peak location does not stabilize early enough to yield an accurate point prediction. In Section 7 we introduce a two-stage construction based on the oscillations of the sequential estimates $T \mapsto \hat{l}(T)$. It produces candidate entry and exit times, and hence candidate peak regions, using only the estimation trajectory; knowledge of the true peak location is not required operationally, and when it is available it is used only retrospectively to evaluate the candidates.

The methodology is applied to daily COVID-19 case counts during the first wave of the pandemic in China Barmparis and Tsironis (2020), France, and Germany, for several quantile levels (Section 8).

The remainder of the paper is organized as follows. In Section 2, we introduce the Gaussian peak model and the quantile-based estimation procedure, including the weighted τ-quantile characterization of the amplitude estimator. Section 3 studies the behavior of the quantile-based objective in the noise-free setting. Section 4 introduces the τ-aligned asymmetric Laplace noise model and establishes its connection with the quantile loss. Section 5 investigates the trajectories of the estimated peak parameters under high signal-to-noise ratio. Section 6 examines the temporal stability of the estimated peak location when the signal-to-noise ratio is reduced. Section 7 introduces the two-stage construction of candidate entry and exit times and the associated peak regions. Finally, Section 8 applies the proposed methodology to daily COVID-19 case data from China, France, and Germany.

2 Quantile-Based Estimation Methodology

We consider the problem of estimating the amplitude, location, and spread of a unimodal signal corrupted by noise. The model is:

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right) + e(t), \qquad t = 1, \ldots, T. \tag{1}$$

where $A > 0$ is the amplitude, $L > 0$ the peak location, $S > 0$ the spread (scale), and $e(t)$ is additive noise. Given observations $(I(t))_{t=1}^{T}$, our goal is to estimate the underlying parameters using quantile regression, i.e., minimizing the asymmetric loss

$$\rho_\tau(u) = u\left(\tau - \mathbf{1}_{\{u<0\}}\right). \tag{2}$$

for a fixed quantile level $\tau \in (0, 1)$. We define the quantile objective:

$$f_\tau(T, a, l, s) = \frac{1}{T}\sum_{t=1}^{T} \rho_\tau\left(I(t) - a \exp\left(-\frac{(t-l)^2}{s^2}\right)\right). \tag{3}$$

For fixed candidate values of (l, s), the optimal amplitude a minimizes $a \mapsto f_\tau(T, a, l, s)$. This minimizer is a weighted τ-quantile of a transformed version of the data, as described below.

2.1 Weighted Quantile Estimation of Amplitude

Let $w(t) = \exp\left(-\frac{(t-l)^2}{s^2}\right)$ be the weight associated with time point t, and define:

$$x(t) = I(t) \exp\left(\frac{(t-l)^2}{s^2}\right).$$

Then the optimal amplitude a is the weighted τ-quantile of $x(t)$ with weights $w(t)$. The following proposition gives the corresponding characterization of the weighted quantile as the minimizer of the weighted quantile loss. The proof of the following Proposition is the same as in the work of Novoselac (2020).

Proposition 1. *Let $(x(t), w(t))$ be a sequence of real values with positive weights $w(t) > 0$. Then, for any $\tau \in (0,1)$, the weighted τ-quantile minimizes the weighted quantile loss:*

$$q_\tau = \arg\min_{a \in \mathbb{R}} \sum_{t=1}^{T} w(t)\, \rho_\tau(a - x(t)),$$

where ρ_τ is the quantile check function defined in (2). To compute q_τ, sort the values $x(t)$ in increasing order, obtaining a permutation π such that

$$x(\pi(1)) \leq x(\pi(2)) \leq \cdots \leq x(\pi(T)).$$

Define the reordered weights $w_\pi(t) = w(\pi(t))$. Let

$$k = \min \left\{ j \in \{1, \ldots, T\} : \sum_{t=1}^{j} w_\pi(t) \geq \tau \sum_{t=1}^{T} w(t) \right\}.$$

Then the weighted τ-quantile satisfies $q_\tau = x(\pi(k))$.

2.2 Quantile-Based Estimation Procedure

We define the following steps to jointly estimate the location $\hat{l}(T)$, amplitude $\hat{a}(T)$, and spread $\hat{s}(T)$:

Algorithm 1 Quantile-based estimation of peak parameters

Require: Time horizon T, upper bound $h > T$, quantile level $\tau \in (0,1)$

1: For each candidate $l \in [1, h]$, find $s^*(l) = \arg\min_s f_\tau(T, a^*(l,s), l, s)$
2: Set $\hat{l}(T) = \arg\min_l f_\tau(T, a^*(l, s^*(l)), l, s^*(l))$
3: Set $\hat{s}(T) = s^*(\hat{l}(T))$
4: Set $\hat{a}(T) = a^*(\hat{l}(T), \hat{s}(T))$

Ensure: Estimated parameters $(\hat{l}(T), \hat{a}(T), \hat{s}(T))$

2.3 Remark on Asymmetry and Robustness

This method generalizes the classic Least Absolute Deviations (LAD) estimator, recovered at $\tau = 0.5$. By adjusting τ, we introduce asymmetry into the estimation loss, enabling the model to better adapt to skewed noise distributions. This makes the approach especially valuable in early-warning systems and epidemic peak detection where overestimation and underestimation have asymmetric consequences.

3 Quantile-Based Objective Function under Noise-Free Conditions

We first consider the idealized scenario where the signal is not corrupted by noise. The model of interest is given by

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right), \quad t = 1, \ldots, T,$$

where $A = 15000$, $L = 22$, $S = 10$, and $T \in \{2, 3, 10, 20, 21\}$. The quantile-based estimation procedure is implemented by minimizing the following objective function (3). The amplitude a is computed via a weighted τ-quantile, as described in Section 2.

Since the goal is to predict the location L of the peak before it becomes observable, we restrict the minimization over candidate peak locations $l > T$. For each value of $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, we compute and visualize the objective curve $l \mapsto f_\tau(T, a(T,l), l, s(T,l))$ for the selected values of T (see Figure 1).

Remark. For each fixed sample size T, we computed the value of the minimized objective function at the estimated parameters $(\hat{a}, \hat{l}, \hat{s})$. The results indicate that the lowest objective value is typically achieved for $\tau = 0.95$, suggesting that upper quantiles align particularly well with the noiseless signal structure. The only exception is at $T = 20$, where $\tau = 0.75$ yields a slightly smaller loss. This empirical observation may help guide quantile selection when prior knowledge of signal shape or noise properties is limited.

4 Quantile-Based Estimation with τ-Aligned Noise

To analyze how quantile-aligned noise affects peak estimation, we introduce a unified noise model based on the *general asymmetric Laplace* (GAL) distribution. This flexible family allows both symmetric and asymmetric noise structures while maintaining the property that the τ-quantile of the noise is at zero.

4.1 Asymmetric Laplace Noise Aligned with the τ-Quantile

We consider an asymmetric Laplace distribution parametrized by a location parameter $m \in \mathbb{R}$, a scale parameter $\lambda > 0$, and an asymmetry parameter $k > 0$. Its density is given by

$$f_{m,\lambda,k}(x) = \frac{\lambda}{k + k^{-1}} \begin{cases} \exp\left(\dfrac{\lambda(x-m)}{k}\right), & x < m, \\ \exp\left(-\lambda k(x-m)\right), & x \geq m. \end{cases}$$

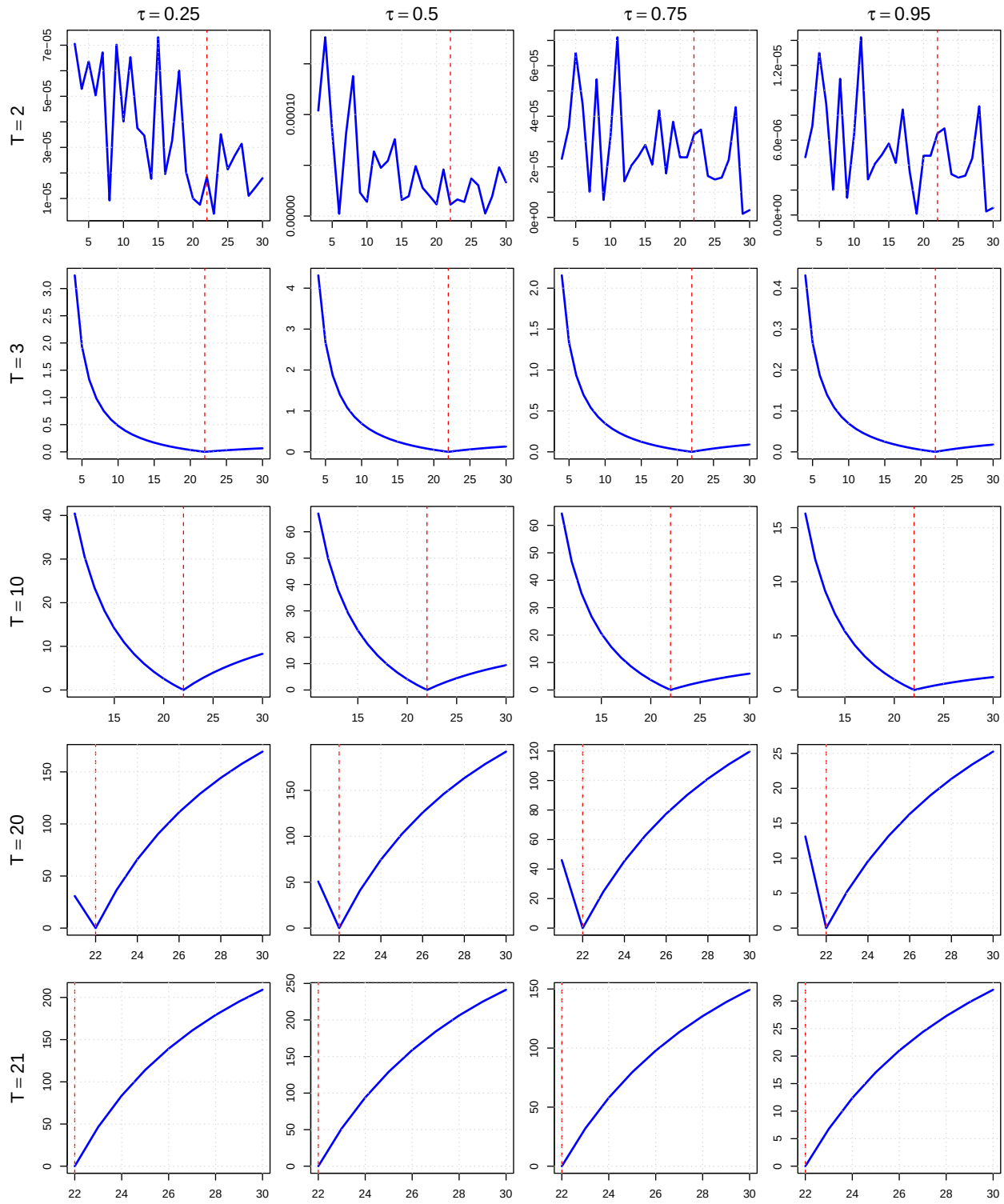


Figure 1: Objective function $l \mapsto f_\tau(T, a, l, s)$ computed for $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and $T \in \{2, 3, 10, 20, 21\}$, under a noise-free signal. The vertical dashed line indicates the true peak location $L = 22$, and the search is restricted to values $l > T$.

The parameter m is the mode of the distribution, while k controls its asymmetry and λ controls its concentration. The distribution function evaluated at the mode is

$$\mathbb{P}(X < m) = \frac{\lambda}{k + k^{-1}} \int_{-\infty}^{m} \exp\left(\frac{\lambda(x-m)}{k}\right) dx = \frac{k^2}{1+k^2}.$$

Therefore, to align the asymmetric Laplace noise with the τ-quantile regression criterion, we require

$$\mathbb{P}(e(t) < 0) = \tau, \tag{4}$$

and $\tau \in (0,1)$. Since the location of the noise can be fixed at zero, we set $m = 0$. The corresponding asymmetry parameter is then determined by

$$\frac{k^2}{1+k^2} = \tau,$$

or equivalently,

$$k = \sqrt{\frac{\tau}{1-\tau}}. \tag{5}$$

Thus, the τ-aligned asymmetric Laplace noise has density

$$f_{\tau,\lambda}(x) = \frac{\lambda}{k + k^{-1}} \begin{cases} \exp\left(\dfrac{\lambda x}{k}\right), & x < 0, \\ \exp(-\lambda k x), & x \geq 0, \end{cases}$$

where k is given by (5). By construction, the noise satisfies the τ-alignment property (4). In particular, when $\tau = 1/2$, we have $k = 1$, and the asymmetric Laplace distribution reduces to the symmetric Laplace distribution centered at zero. Hence, the symmetric Laplace case is recovered as a special case of the τ-aligned asymmetric Laplace model.

4.2 Simulation under Asymmetric Laplace Noise

We now investigate the behavior of the quantile-based peak estimation procedure when the additive noise follows the τ-aligned asymmetric Laplace distribution introduced in Section 4.1. In the parameterization based on the quantile loss, the density of the noise can be written as

$$f_{\tau,b}(x) = \frac{\tau(1-\tau)}{b} \exp\left(-\frac{1}{b}\rho_\tau(x)\right), \qquad \tau \in (0,1), \quad b > 0. \tag{6}$$

where ρ_τ is the check function defined in (2). Equivalently,

$$f_{\tau,b}(x) = \begin{cases} \dfrac{\tau(1-\tau)}{b} \exp\left(-\dfrac{\tau x}{b}\right), & x \geq 0, \\ \dfrac{\tau(1-\tau)}{b} \exp\left(\dfrac{(1-\tau)x}{b}\right), & x < 0. \end{cases}$$

The normalization follows from

$$\int_{-\infty}^{+\infty} \exp\left(-\frac{1}{b}\rho_\tau(x)\right) dx = \frac{b}{1-\tau} + \frac{b}{\tau} = \frac{b}{\tau(1-\tau)}.$$

Moreover, by construction, the noise satisfies the τ-alignment property (4). This distribution is directly compatible with the quantile-based estimation criterion. Indeed,

$$-\log f_{\tau,b}(e) = -\log\left(\frac{\tau(1-\tau)}{b}\right) + \frac{1}{b}\rho_\tau(e).$$

Therefore, for fixed τ and b, minimizing the negative log-likelihood is equivalent to minimizing the empirical quantile loss. The asymmetric Laplace model thus provides a probabilistic interpretation of the quantile-based peak estimation procedure without introducing a separate estimation criterion. For the simulation study, the true signal is

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right),$$

with $A = 1$, $L = 22$, $S = 10$. Since the maximum amplitude of the Gaussian signal is $\max_t I(t) = A$, we define the signal-to-noise ratio as the ratio between the peak amplitude of the signal and the standard deviation of the noise:

$$\mathrm{SNR} = \frac{A}{\sqrt{\mathrm{Var}(e(t))}}. \tag{7}$$

In the simulations, we set SNR = 1000. For each $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, the corresponding asymmetric Laplace noise is generated with asymmetry parameter (5). Thus, the noise variance is calibrated according to

$$\mathrm{Var}(e(t)) = \left(\frac{A}{\mathrm{SNR}}\right)^2. \tag{8}$$

For the τ-aligned asymmetric Laplace distribution, this variance is

$$\mathrm{Var}(e(t)) = b^2 \frac{\tau^2 + (1-\tau)^2}{\tau^2(1-\tau)^2}. \tag{9}$$

Combining (8) and (9) gives

$$b = \frac{A}{\mathrm{SNR}} \frac{\tau(1-\tau)}{\sqrt{\tau^2 + (1-\tau)^2}}. \tag{10}$$

For each τ, the objective function $l \mapsto f_\tau(T, a, l, s)$ is evaluated for the time horizons $T \in \{4, 6, 8, 9, 10\}$. For each candidate peak location l, the scale parameter s is optimized over the prescribed interval, and the amplitude a is determined as the weighted τ-quantile described in Section 2.1. The resulting objective profiles are then compared across the different values of τ. The resulting profiles are displayed in Figure 2. The true peak location $L = 22$ is marked by a vertical dashed line. Each panel corresponds to a different pair (T, τ), and the curves show the behavior of the quantile-loss objective across candidate peak locations.

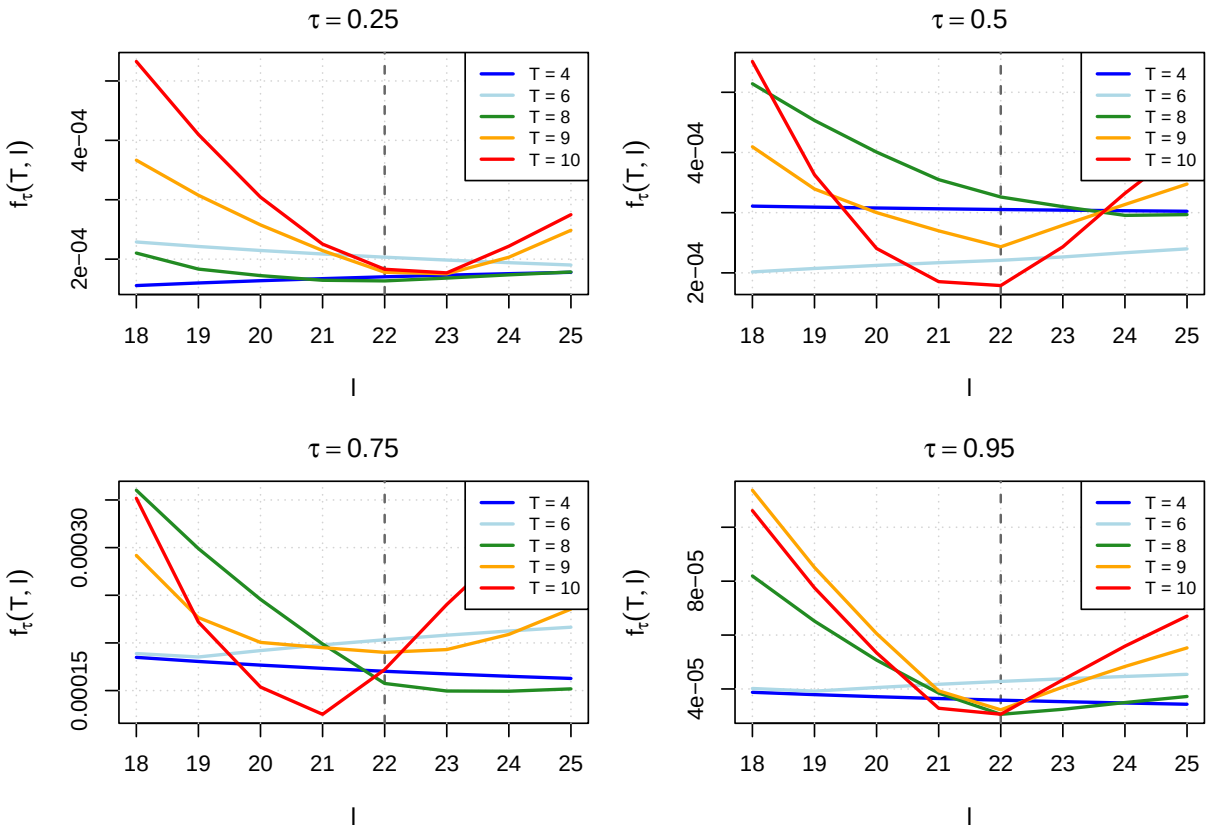


Figure 2: Objective function profiles $l \mapsto f_\tau(T, a, l, s)$ under *asymmetric Laplace* noise, for $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and $T \in \{4, 6, 8, 9, 10\}$. Each subfigure corresponds to one value of τ, and the dashed vertical line indicates the true peak location $L = 22$.

5 Parameter Estimation Trajectories

We now analyze how the estimated model parameters, location $\hat{l}(T)$, amplitude $\hat{a}(T)$, and scale $\hat{s}(T)$, evolve as the sample size T increases. Our goal is to assess the stability and convergence of these estimators under the τ-aligned asymmetric Laplace noise introduced in Section 4.1, including its symmetric special case. All experiments use the SNR definition given in (7), with SNR $= 1000$, and consider quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$.

5.1 τ-Aligned Asymmetric Laplace Noise

We consider the signal-plus-noise model (1), with $T = 30$, $A = 1$, $L = 22$, $S = 10$. The noise $e(t)$ follows the τ-aligned asymmetric Laplace distribution introduced in Section 4.1, with density (6). The asymmetry parameter is determined by the quantile level according to (5), so that the τ-alignment property (4) holds. In particular, $\tau = 1/2$ corresponds to the symmetric Laplace distribution, whereas the other values of τ considered here produce asymmetric noise. The scale parameter b is calibrated separately for each τ so that the variance of the noise satisfies the fixed signal-to-noise ratio condition (8). For the τ-aligned asymmetric Laplace distribution, this gives (9), and therefore (10). At each time horizon $T \in \{4, \ldots, 30\}$, we estimate the parameters $(\hat{l}(T), \hat{a}(T), \hat{s}(T))$ by minimizing the objective function (3) over the candidate peak locations and scale parameters, with the amplitude a re-estimated for each pair (l, s) as the weighted τ-quantile described in Section 2.1. The scale parameter s is optimized over the prescribed interval, and the location parameter l is selected by minimizing the resulting objective over the candidate locations. Figure 3 shows the resulting estimation trajectories $T \mapsto \hat{l}(T)$, $T \mapsto \hat{a}(T)$, $T \mapsto \hat{s}(T)$, for each quantile level τ. The true parameter values $(L, A, S) = (22, 1, 10)$ are indicated by dashed horizontal lines.

6 Stability Analysis

This section investigates the temporal stability of the estimated peak location when the signal-to-noise ratio is reduced. We focus on the τ-aligned asymmetric Laplace noise introduced in Section 4.1 and examine how the estimated peak location evolves as new observations become available. In particular, we compare the stability of the estimation paths across different quantile levels and assess the effect of a lower signal-to-noise ratio on the ability to recover the true peak location.

6.1 Low Signal-to-Noise Performance

To evaluate the robustness of the quantile-based estimator under a lower signal-to-noise ratio, we consider the signal-plus-noise model (1), with $T = 35$, $A = 1$, $L = 22$, $S = 10$. The noise $e(t)$ follows the τ-aligned asymmetric Laplace distribution introduced in Section 4.1, with density (6). For each $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, the scale parameter b is calibrated so

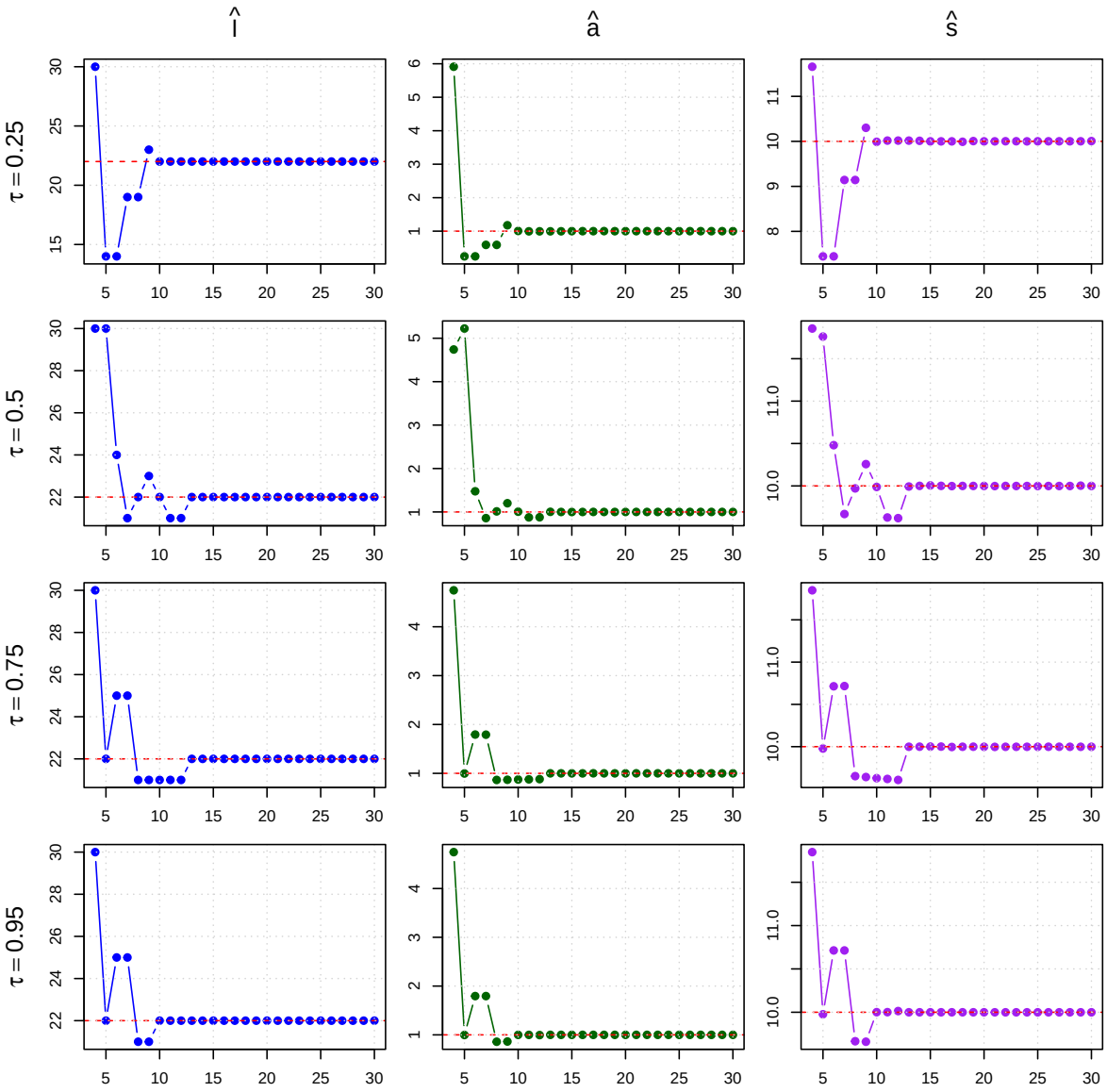


Figure 3: Estimation paths $T \mapsto \hat{l}(T)$ (left), $\hat{a}(T)$ (middle), and $\hat{s}(T)$ (right) under τ-aligned asymmetric Laplace noise with signal-to-noise ratio (SNR) 1000. Each row corresponds to a quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$. The true values $(L, A, S) = (22, 1, 10)$ are indicated by dashed horizontal lines.

that (8), with SNR = 40, according to the definition in (7). Using the variance of the τ-aligned asymmetric Laplace distribution gives (10). For each quantile level τ, we compute the estimation path $T \mapsto \hat{l}(T)$, $T = 4, \ldots, 35$, where $(\hat{l}(T), \hat{a}(T), \hat{s}(T))$ is obtained by minimizing the empirical quantile loss over the candidate peak locations and scale parameters, with the amplitude re-estimated as the weighted τ-quantile described in Section 2.1. To characterize the temporal stability of the estimated peak location, we consider the earliest observation horizon T_{stab} from which all subsequent estimates remain within a prescribed range of one another. Specifically, for a tolerance $\varepsilon = 4$, we define

$$T_{\text{stab}} = \min \left\{ T : \max_{u \geq T} \hat{l}(u) - \min_{u \geq T} \hat{l}(u) \leq \varepsilon \right\}.$$

This stabilization time measures the earliest point from which the remaining estimation path remains confined to a stable range. Figure 4 displays the resulting estimation paths for the four quantile levels. The horizontal dashed line indicates the true peak location $L = 22$. The comparison illustrates how the temporal stability of the estimated peak location depends on the quantile level when the signal-to-noise ratio is reduced. The corresponding stabilization times are reported in Table 1. Smaller values of T_{stab} indicate that the estimator reaches a stable regime earlier, whereas larger values indicate that additional observations are required before the estimated peak location becomes stable.

Table 1: Stabilization times T_{stab} for the estimated peak location under SNR = 40 and different quantile levels.

	$\tau = 0.25$	$\tau = 0.5$	$\tau = 0.75$	$\tau = 0.95$
T_{stab}	17	17	11	10

7 Entry and Exit Times of the Peak Region

The sequential estimation procedure introduced in the preceding sections provides, for each observation horizon T, an estimate $\hat{l}(T)$ of the peak location. The stabilization criterion considered in Section 6 characterizes the point from which the future estimation path remains within a prescribed tolerance. In this section, we introduce a complementary construction based on the oscillations of the estimated peak location. This construction aims at identifying candidate time intervals describing the region in which the estimated peak location becomes stable, through a data-driven time of entry and a time of exit. The operational construction is based only on the sequential estimation trajectory $T \longmapsto \hat{l}(T)$. The true peak location L is not required to construct the candidate entry and exit times. In the simulation study, where L is known, it is used only for an ex post assessment of whether a candidate peak region contains the true peak.

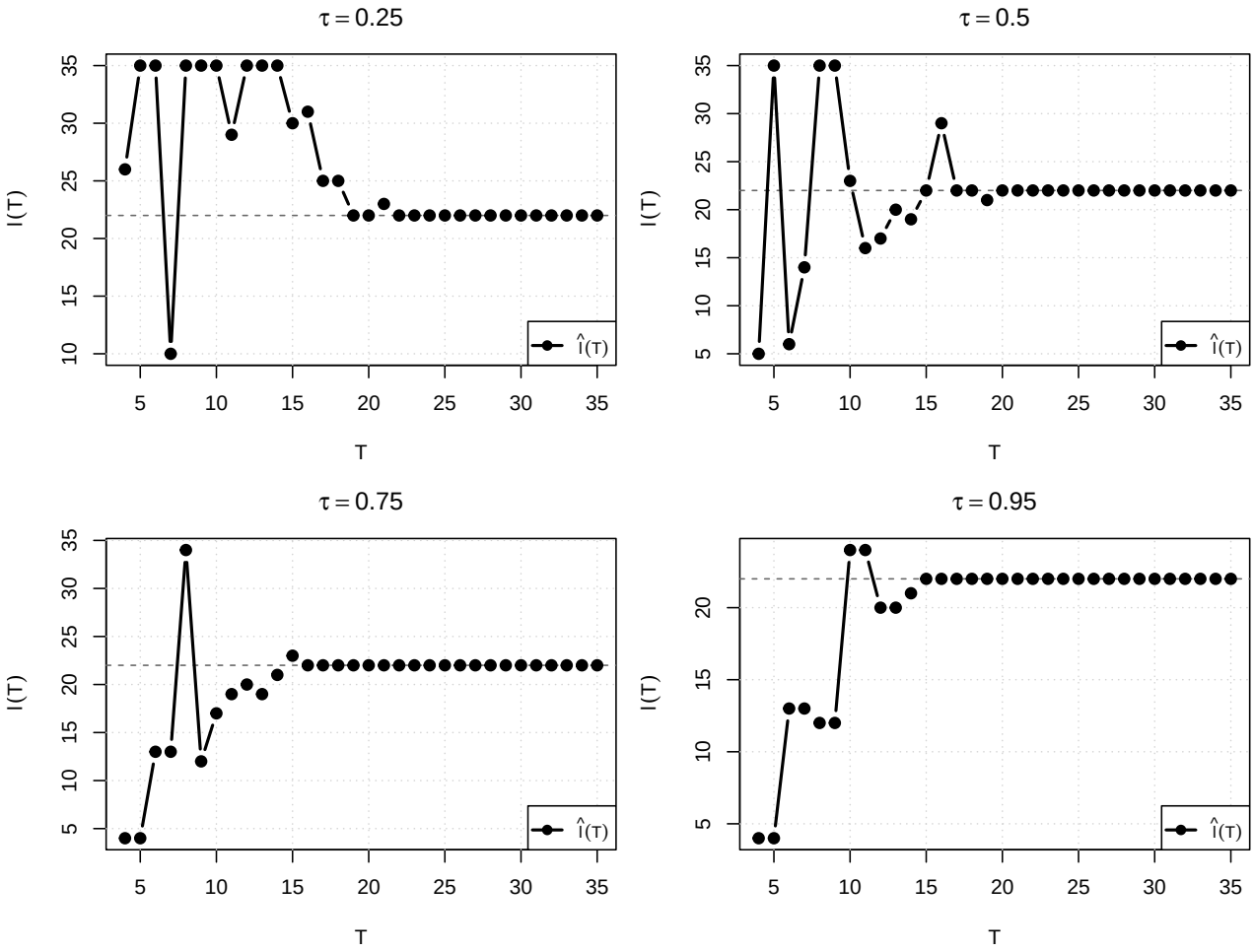


Figure 4: Estimation paths $T \mapsto \hat{l}(T)$ under a low signal-to-noise ratio (SNR $= 40$), for quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$. The horizontal dashed line indicates the true peak location $L = 22$.

7.1 Oscillations of the estimated peak location

Let $T_0 < T_0 + 1 < \cdots < h$ denote the observation horizons for which the sequential estimator $\hat{l}(T)$ is available. For two observation times T and T' satisfying $T_0 \le T < T' \le h$, we define the oscillation of the estimated peak location over the interval $[T, T']$ by

$$\omega_l(T, T') = \max_{T \le t_1, t_2 \le T'} \left| \hat{l}(t_1) - \hat{l}(t_2) \right|. \tag{11}$$

Thus, $\omega_l(T, T')$ measures the largest variation of the estimated peak location over the interval $[T, T']$. In particular, $\omega_l(T, T') = 0$ if and only if $\hat{l}(t)$ is constant over the observation horizons contained in $[T, T']$. A particular case of (11) is the future-tail oscillation

$$\omega_l(T, h) = \max_{T \le t_1, t_2 \le h} \left| \hat{l}(t_1) - \hat{l}(t_2) \right|.$$

This quantity measures the largest variation that can still occur in the estimated peak location after observation time T. The future-tail oscillation is nonincreasing with T. Indeed, if $T_1 < T_2 \le h$, then $[T_2, h] \subset [T_1, h]$, and consequently $\omega_l(T_2, h) \le \omega_l(T_1, h)$. Hence, the function $g(T) = \omega_l(T, h)$ provides a nonincreasing measure of the remaining temporal variation of the estimated peak location.

7.2 Recursive construction of candidate exit times

The candidate exit times are constructed recursively from the future-tail oscillation. Let $h = \max\{T : \hat{l}(T)\}$ is available. The initial candidate is defined by $\tau'_0 = 0$, $T^{(0)}_{\text{exit}} = h$. We also initialize the current boundary at $B_0 = h$. Suppose that at a given step k we have a current oscillation level τ'_k and a current boundary B_k. We consider observation times strictly before the current boundary and search for the earliest observation time T'_k satisfying

$$T'_k < B_k, \qquad |\omega_l(T'_k, h) - \tau'_k| \le \varepsilon_1, \tag{12}$$

where ε_1 is a numerical tolerance used to implement the equality in the R procedure. If such an observation time exists, let $T^{(k)}_{\text{prev}} = \max\{T < T'_k\}$ be the observation horizon immediately preceding T'_k. The next candidate exit time and oscillation level are then defined by $T^{(k+1)}_{\text{exit}} = T^{(k)}_{\text{prev}}$ and $\tau'_{k+1} = \omega_l\left(T^{(k)}_{\text{prev}}, h\right)$. The new boundary is set to $B_{k+1} = T'_k$. If no observation time satisfying (12) exists before the current boundary, the recursion does not terminate immediately. Instead, the observation horizon immediately preceding the current boundary is taken as the next candidate exit time: $T^{(k+1)}_{\text{exit}} = \max\{T < B_k\}$, with corresponding oscillation level $\tau'_{k+1} = \omega_l\left(T^{(k+1)}_{\text{exit}}, h\right)$, and the new boundary is set to $B_{k+1} = T^{(k+1)}_{\text{exit}}$. The recursion is continued as long as the next oscillation level increases numerically. In the implementation, the condition $\tau'_{k+1} > \tau'_k$ is checked up to a numerical tolerance ε_1. This recursive construction generates a sequence $\left\{\left(\tau'_k, T^{(k)}_{\text{exit}}\right)\right\}_{k=0}^{K}$, where each pair consists of a candidate exit time and the corresponding future-tail oscillation. Importantly, the construction uses only the observed trajectory $\hat{l}(T)$ and

does not require knowledge of the true peak location L.

7.3 Determination of the entry time

For a given candidate exit time $T_{\text{exit}}^{(k)}$, we first consider all observation times strictly before the candidate exit time. The set

$$E_k = \left\{ T < T_{\text{exit}}^{(k)} : \omega_l(T, T_{\text{exit}}^{(k)}) \geq \tau_k' - \varepsilon_2 \right\},$$

is the numerical implementation of the Stage-2 set, where ε_2 is a numerical tolerance. If E_k is empty, the candidate does not yield a valid entry time. Otherwise, we define $T_k^* = \max E_k$. The corresponding oscillation level is $\tau_k = \omega_l \left(T_k^*, T_{\text{exit}}^{(k)} \right)$. We then consider all observation times $T < T_{\text{exit}}^{(k)}$ for which $\left| \omega_l(T, T_{\text{exit}}^{(k)}) - \tau_k \right| \leq \varepsilon_3$, where ε_3 is the numerical tolerance used for equality in Stage 2. The entry time is defined as the first such observation horizon:

$$T_{\text{entry}}^{(k)} = \min \left\{ T < T_{\text{exit}}^{(k)} : \left| \omega_l(T, T_{\text{exit}}^{(k)}) - \tau_k \right| \leq \varepsilon_3 \right\},$$

provided that such an observation time exists. The width associated with the candidate is

$$w_k = T_{\text{exit}}^{(k)} - T_{\text{entry}}^{(k)}. \tag{13}$$

The resulting candidate peak region is therefore represented by the interval $\mathcal{R}_k = \left[T_{\text{entry}}^{(k)}, T_{\text{exit}}^{(k)} \right]$. The endpoints are determined entirely from the sequential estimation trajectory and the oscillation construction.

8 Real-World Application: COVID-19 Case Study

We now demonstrate the application of our quantile-based peak estimation framework to real-world data. We focus on the daily confirmed COVID-19 cases in China, France, and Germany. Figure 5 displays the daily case series for these three countries. For each country, we analyze the sequence $T \mapsto \hat{l}(T)$ for the quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, using quantile-loss minimization over the Gaussian-shaped parametric peak model. Following Algorithm 1 of the manuscript, the candidate peak location ranges over the analysis horizon $h = \max(T)$, so that for each country the search set is $\{1, \ldots, h\}$. We then assess the temporal stabilization of the estimated peak-location path using the retrospective stabilization criterion introduced in Section 6, applied separately to each country and quantile level; the resulting stabilization time is denoted T_{stab}. For each country we additionally apply the operational two-stage entry–exit construction of Section 7, which uses the oscillations of the estimated peak-location path to identify a candidate peak-region window $(T_{\text{entry}}^{(k)}, T_{\text{exit}}^{(k)})$ and whose width is denoted $w_k = T_{\text{exit}}^{(k)} - T_{\text{entry}}^{(k)}$. Because the observed peak location L is known for each of the three country-level trajectories used here, the resulting Stage-2 candidates can be evaluated retrospectively, and the narrowest candidate satisfying the strict containment criterion $T_{\text{entry}}^{(k)} < L < T_{\text{exit}}^{(k)}$ is

reported for each country and quantile level. This minimum-width selection is used only for retrospective evaluation and is not part of the operational procedure when L is unknown.

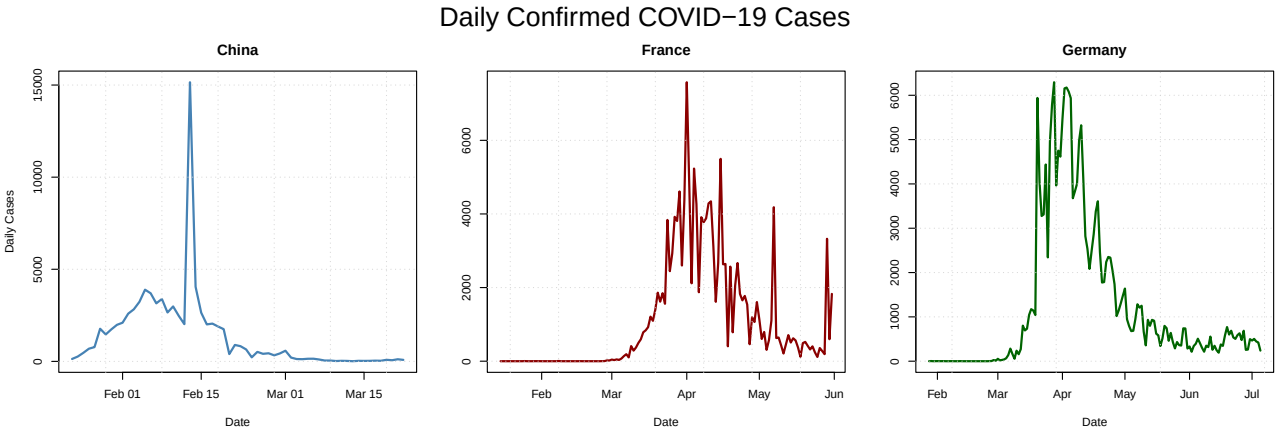


Figure 5: Daily confirmed COVID-19 cases in China, France, and Germany (left to right).

8.1 China

We analyze the first wave of the COVID-19 epidemic in China, using the first 60 daily new case counts from January 22 to March 22, 2020. The estimation is performed for observation horizons $T = 4, \ldots, 35$. For each quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, we compute the sequence $T \mapsto \hat{l}(T)$ by minimizing the empirical quantile loss over the Gaussian-shaped peak model, with candidate peak locations ranging over $l \in \{1, \ldots, h\}$ with $h = 35$. Temporal stabilization is assessed using the criterion introduced in Section 6, with tolerance $\varepsilon = 3$. The resulting estimated peak-location sequences are reported in Table 5.

The corresponding stabilization times are $T_{\text{stab}} = 15$, 13, 23, and 28 for $\tau = 0.25$, 0.5, 0.75, and 0.95, respectively. The stabilization criterion characterizes the temporal behavior of the estimated peak-location path and is distinct from the entry–exit construction introduced in Section 7. For the present real-data application, where the observed peak location is known retrospectively, we apply the latter construction and, among the Stage-2 candidates satisfying $T_{\text{entry}}^{(k)} < L < T_{\text{exit}}^{(k)}$, retain the one with the smallest width $w_k = T_{\text{exit}}^{(k)} - T_{\text{entry}}^{(k)}$. This minimum-width selection is used only for retrospective evaluation and is not part of the operational procedure when the peak location is unknown.

For China, the observed peak location is $L = 22$. The resulting retrospective selection for the four quantile levels is reported in Table 2. For $\tau = 0.25$, the selected candidate corresponds to $\tau_k' = 1$, with $T_{\text{exit}} = 26$, $T^* = 15$, $\tau = 3$, and $T_{\text{entry}} = 15$, giving a window width $w^* = 11$. For $\tau = 0.5$, the selected candidate corresponds to $\tau_k' = 2$, with $T_{\text{exit}} = 24$, $T^* = 15$, $\tau = 2$, and $T_{\text{entry}} = 15$, giving $w^* = 9$. For $\tau = 0.95$, the selected candidate corresponds to $\tau_k' = 15$, with $T_{\text{exit}} = 26$, $T^* = 21$, $\tau = 21$, and $T_{\text{entry}} = 13$, giving $w^* = 13$. For $\tau = 0.75$, no Stage-2 candidate satisfies the strict containment condition $T_{\text{entry}}^{(k)} < 22 < T_{\text{exit}}^{(k)}$, and consequently no retrospective candidate is selected.

The results show that the retrospective peak-region candidate depends on the quantile level.

Table 2: Retrospective selection of the peak-region candidate for COVID-19 in China.

τ	k	τ'_k	T_{exit}	T^*	τ	T_{entry}	w^*	L
0.25	1	1	26	15	3	15	11	22
0.50	2	2	24	15	2	15	9	22
0.75	–	–	–	–	–	–	–	22
0.95	4	15	26	21	21	13	13	22

The selected windows are $(15, 26)$ for $\tau = 0.25$, $(15, 24)$ for $\tau = 0.5$, and $(13, 26)$ for $\tau = 0.95$. For $\tau = 0.75$, the two-stage construction does not produce a candidate whose window strictly contains the observed peak location $L = 22$. These results complement the stabilization analysis in Table 5: the stabilization time describes the temporal stabilization of the estimated peak-location path, whereas the two-stage construction uses its oscillations to identify candidate windows for the peak region.

8.2 France

Next, we apply the estimation method to daily COVID-19 cases in France, using ECDC data from January 15 to May 31, 2020. The analysis covers observation horizons $T = 50, \ldots, 89$. For each quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, we compute the corresponding estimation path $T \mapsto \hat{l}(T)$, with candidate peak locations ranging over $l \in \{1, \ldots, h\}$ with $h = \max(T) = 89$. Temporal stabilization is assessed using the criterion introduced in Section 6, with tolerance $\varepsilon = 5$. The estimated peak-location sequences are reported in Table 6.

The corresponding stabilization times are $T_{\text{stab}} = 85$, 82, 80, and 85 for $\tau = 0.25$, 0.5, 0.75, and 0.95, respectively. The stabilization criterion characterizes the temporal behavior of the estimated peak-location path and is distinct from the entry–exit construction introduced in Section 7. For the present real-data application, where the observed peak location is known retrospectively, we apply the latter construction and, among the Stage-2 candidates satisfying $T^{(k)}_{\text{entry}} < L < T^{(k)}_{\text{exit}}$, retain the one with the smallest width $w_k = T^{(k)}_{\text{exit}} - T^{(k)}_{\text{entry}}$. This minimum-width selection is used only for retrospective evaluation and is not part of the operational procedure when the peak location is unknown.

For France, the observed peak location is $L = 78$. The resulting retrospective selection for the four quantile levels is reported in Table 3. For $\tau = 0.25$, the selected candidate corresponds to $\tau'_k = 13$, with $T_{\text{exit}} = 79$, $T^* = 76$, $\tau = 13$, and $T_{\text{entry}} = 72$, giving a window width $w^* = 7$. For $\tau = 0.5$, the selected candidate corresponds to $\tau'_k = 4$, with $T_{\text{exit}} = 82$, $T^* = 78$, $\tau = 6$, and $T_{\text{entry}} = 73$, giving $w^* = 9$. For $\tau = 0.75$, the selected candidate corresponds to $\tau'_k = 8$, with $T_{\text{exit}} = 79$, $T^* = 72$, $\tau = 11$, and $T_{\text{entry}} = 70$, giving $w^* = 9$. For $\tau = 0.95$, the selected candidate corresponds to $\tau'_k = 11$, with $T_{\text{exit}} = 80$, $T^* = 77$, $\tau = 13$, and $T_{\text{entry}} = 70$, giving $w^* = 10$.

The results show that the retrospective peak-region candidate depends on the quantile level. The selected windows are $(72, 79)$ for $\tau = 0.25$, $(73, 82)$ for $\tau = 0.5$, $(70, 79)$ for $\tau = 0.75$, and $(70, 80)$ for $\tau = 0.95$. All four windows strictly contain the observed peak location $L = 78$, so

Table 3: Retrospective selection of the peak-region candidate for COVID-19 in France.

τ	k	τ'_k	T_{exit}	T^*	τ	T_{entry}	w^*	L
0.25	5	13	79	76	13	72	7	78
0.50	3	4	82	78	6	73	9	78
0.75	3	8	79	72	11	70	9	78
0.95	6	11	80	77	13	70	10	78

a retrospective candidate is selected for every quantile level; the widths vary between 7 and 10 across the four quantile specifications. These results complement the stabilization analysis in Table 6: the stabilization time describes the temporal stabilization of the estimated peak-location path, whereas the two-stage construction uses its oscillations to identify candidate windows for the peak region.

8.3 Germany

Finally, we consider COVID-19 daily cases in Germany from January 28 to July 5, 2020. To align the peak location with day $t = 61$, we index the data starting on January 28. We analyze observation horizons $T = 40, \ldots, 75$. For each quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, we compute the estimation path $T \mapsto \hat{l}(T)$, with candidate peak locations ranging over $l \in \{1, \ldots, h\}$ with $h = 75$. Temporal stabilization is assessed using the criterion introduced in Section 6, with tolerance $\varepsilon = 15$. The resulting estimation paths are reported in Table 7.

The corresponding stabilization times are $T_{\text{stab}} = 60$, 59, 56, and 57 for $\tau = 0.25$, 0.5, 0.75, and 0.95, respectively. The stabilization criterion characterizes the temporal behavior of the estimated peak-location path and is distinct from the entry–exit construction introduced in Section 7. For the present real-data application, where the observed peak location is known retrospectively, we apply the latter construction and, among the Stage-2 candidates satisfying $T^{(k)}_{\text{entry}} < L < T^{(k)}_{\text{exit}}$, retain the one with the smallest width $w_k = T^{(k)}_{\text{exit}} - T^{(k)}_{\text{entry}}$. This minimum-width selection is used only for retrospective evaluation and is not part of the operational procedure when the peak location is unknown.

For Germany, the observed peak location is $L = 61$. The resulting retrospective selection for the four quantile levels is reported in Table 4. For $\tau = 0.25$, the selected candidate corresponds to $\tau'_k = 15$, with $T_{\text{exit}} = 62$, $T^* = 60$, $\tau = 15$, and $T_{\text{entry}} = 60$, giving a window width $w^* = 2$. For $\tau = 0.95$, the selected candidate corresponds to $\tau'_k = 9$, with $T_{\text{exit}} = 63$, $T^* = 56$, $\tau = 18$, and $T_{\text{entry}} = 56$, giving $w^* = 7$. For $\tau = 0.5$ and $\tau = 0.75$, no Stage-2 candidate satisfies the strict containment condition $T^{(k)}_{\text{entry}} < 61 < T^{(k)}_{\text{exit}}$, and consequently no retrospective candidate is selected.

The results show that the retrospective peak-region candidate depends on the quantile level. The selected windows are $(60, 62)$ for $\tau = 0.25$, and $(56, 63)$ for $\tau = 0.95$. For $\tau = 0.5$ and $\tau = 0.75$, the two-stage construction does not produce a candidate whose window strictly contains the observed peak location $L = 61$. These results complement the stabilization analysis in Table 7: the stabilization time describes the temporal stabilization of the estimated peak-

Table 4: Retrospective selection of the peak-region candidate for COVID-19 in Germany.

τ	k	τ'_k	T_{exit}	T^*	τ	T_{entry}	w^*	L
0.25	7	15	62	60	15	60	2	61
0.50	–	–	–	–	–	–	–	61
0.75	–	–	–	–	–	–	–	61
0.95	6	9	63	56	18	56	7	61

location path, whereas the two-stage construction uses its oscillations to identify candidate windows for the peak region.

8.4 Comparison Across Quantile Levels

To compare the four quantile specifications, we evaluate the minimized quantile-loss objective at the final observation horizon $T_{\max}$ for each country. This provides a descriptive comparison of the empirical objective values obtained under the different quantile levels considered. The quantile level yielding the smallest objective value at $T_{\max}$ is reported for each data set. The resulting minima are obtained for

- China: $\tau = 0.25$ at $T = 35$,
- France: $\tau = 0.95$ at $T = 89$,
- Germany: $\tau = 0.95$ at $T = 75$.

These results indicate that the empirical quantile-loss criterion favors different quantile specifications across the three epidemic trajectories. In particular, the lower quantile $\tau = 0.25$ yields the smallest objective value for China, whereas the upper quantile $\tau = 0.95$ yields the smallest objective value for both France and Germany. This comparison is descriptive and is intended to characterize the behavior of the proposed estimator under the four quantile levels considered, rather than to establish that any one of these quantiles represents the true conditional quantile of the epidemic process.

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A Trajectory Tables

Table 5: Estimated peak location $\hat{l}(T)$ for COVID-19 in China using quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$.

$\tau = 0.25,\ T_{\text{stab}} = 15,\ \varepsilon = 3$																
T	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$\hat{l}(T)$	6	5	35	14	11	11	11	11	14	14	21	18	15	15	15	15
T	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
$\hat{l}(T)$	15	15	15	15	15	15	15	16	16	16	16	16	16	16	16	16
$\tau = 0.5,\ T_{\text{stab}} = 13,\ \varepsilon = 3$																
T	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$\hat{l}(T)$	6	5	35	8	9	11	11	12	14	15	18	17	16	16	15	16
T	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
$\hat{l}(T)$	15	15	15	16	16	17	17	18	18	18	17	17	17	17	17	17
$\tau = 0.75,\ T_{\text{stab}} = 23,\ \varepsilon = 3$																
T	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$\hat{l}(T)$	6	5	35	8	7	8	10	13	14	16	24	18	17	16	15	15
T	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
$\hat{l}(T)$	15	15	35	20	20	20	18	18	18	18	18	18	18	18	18	18
$\tau = 0.95,\ T_{\text{stab}} = 28,\ \varepsilon = 3$																
T	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$\hat{l}(T)$	6	5	35	9	7	8	9	10	11	15	29	29	15	14	14	15
T	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
$\hat{l}(T)$	15	14	35	35	35	35	35	26	22	21	21	20	20	20	20	20

Table 6: Estimated peak location $\hat{l}(T)$ for COVID-19 in France using quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$.

$\tau = 0.25,\ T_{\text{stab}} = 85,\ \varepsilon = 5$																			
T	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69
$\hat{l}(T)$	63	61	89	89	56	55	89	71	83	71	72	67	65	66	66	66	74	72	72	69
T	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89
$\hat{l}(T)$	69	72	81	82	89	89	76	80	89	88	75	79	83	76	76	80	82	82	82	82
$\tau = 0.5,\ T_{\text{stab}} = 82,\ \varepsilon = 5$																			
T	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69
$\hat{l}(T)$	49	54	88	72	54	89	58	60	89	87	89	71	69	77	67	72	73	71	73	70
T	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89
$\hat{l}(T)$	74	78	78	83	88	89	88	85	89	86	85	86	83	80	79	79	80	80	81	80
$\tau = 0.75,\ T_{\text{stab}} = 80,\ \varepsilon = 5$																			
T	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69
$\hat{l}(T)$	48	51	89	89	89	89	89	59	64	65	87	68	66	81	73	70	83	83	72	72
T	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89
$\hat{l}(T)$	89	89	78	84	84	86	85	83	89	87	83	81	81	81	80	79	79	81	81	81
$\tau = 0.95,\ T_{\text{stab}} = 85,\ \varepsilon = 5$																			
T	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69
$\hat{l}(T)$	48	51	56	89	59	89	89	69	59	60	63	63	64	67	67	70	89	87	77	70
T	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89
$\hat{l}(T)$	89	89	89	89	79	76	76	76	89	89	89	86	86	86	85	83	80	78	79	79

Table 7: Estimated peak location $\hat{l}(T)$ for COVID-19 in Germany using quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$.

$\tau = 0.25,\ T_{\text{stab}} = 60,\ \varepsilon = 15$																		
T	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57
$\hat{l}(T)$	75	45	42	41	44	47	70	75	48	75	54	52	51	51	75	75	75	75
T	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
$\hat{l}(T)$	56	56	75	71	60	61	62	62	68	75	72	70	65	65	65	66	66	66
$\tau = 0.5,\ T_{\text{stab}} = 59,\ \varepsilon = 15$																		
T	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57
$\hat{l}(T)$	75	40	40	46	42	46	75	75	64	74	68	58	51	75	75	75	73	72
T	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
$\hat{l}(T)$	56	63	74	71	62	61	61	61	63	65	66	66	66	64	64	65	66	66
$\tau = 0.75,\ T_{\text{stab}} = 56,\ \varepsilon = 15$																		
T	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57
$\hat{l}(T)$	75	40	40	44	42	47	75	75	54	75	60	57	52	75	75	75	55	56
T	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
$\hat{l}(T)$	56	61	66	67	66	64	60	62	64	64	65	65	65	65	64	65	66	66
$\tau = 0.95,\ T_{\text{stab}} = 57,\ \varepsilon = 15$																		
T	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57
$\hat{l}(T)$	74	50	43	41	41	43	53	53	53	53	54	54	54	75	75	75	71	55
T	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
$\hat{l}(T)$	53	55	57	58	58	57	58	59	60	61	61	61	61	61	61	64	66	66